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TOPIC:
ONLINE HOSTEL MANAGEMENT SYSTEM

CASE STUDY:
PEARLS OF WISDOM HOSTEL

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1. Introduction

Pearls of Wisdom Hostel, located in Kasindikwa village in Fort Portal, Western Uganda, represents an important and growing pillar of student accommodation in the region. Pearls of Wisdom Hostel serves undergraduate and postgraduate students pursuing programs in computer science, education, agriculture, tourism, health sciences, and other disciplines. The hostel offers diverse, affordable, and student-centered accommodation options ranging from basic shared rooms to modern single rooms, double-occupancy units, and fully self-contained rooms.

Pearls of Wisdom Hostel has gained strong popularity among Mountains of the moon university students because of its strategic location, competitive semester rates and focus on comfort. The facility features high-speed Wi-Fi, reliable water supply with storage tanks, study spaces, laundry areas, and proximity to shopping centres, restaurants, and public transport. The hostel management continues to improve its services to meet the expectations of modern students who seek safe, convenient, and conducive living environments close to lectures, the library, and university administration. As a result, Pearls of Wisdom Hostel has become a preferred accommodation choice for both local and out-of-town students, playing a critical role in supporting academic success and overall student welfare in the Tooro region.

2. Background

At Pearls of Wisdom Hostel, accommodation booking process is largely manual and counter-based. Prospective or returning students typically visit the hostel premises in person, especially during peak periods such as the beginning of semesters or after holidays.

Upon arrival, students approach the hostel office or manager and inquire about available rooms. They specify their preferences such as room type (single, double, or self-contained), price range, and intended move-in date. The hostel manager checks room availability by consulting a physical ledger book.

If a suitable room is available, the manager informs the student of the rental cost, which is usually charged per semester. In some cases, the manager may negotiate discounts for early bookings or referrals. Payments are mainly made in cash, although mobile money services such as MTN MoMo or Airtel Money are sometimes accepted.

After payment is made, the student is issued a handwritten receipt or simple agreement note showing the room number, payment details, and move-in date. During check-in, the hostel

warden verifies the receipt, confirms the student's details, and provides the physical room key. Students may also be required to submit passport photos, identification documents, or copies of their student ID cards for record-keeping purposes.

Ongoing hostel management activities such as tracking rent payments, managing room occupancy, and handling maintenance requests are done manually. Payment reminders are usually communicated through phone calls, text messages, or WhatsApp, while maintenance issues are reported verbally to the hostel management.

Challenges in Service Delivery at Pearls of Wisdom Hostel

Despite providing essential accommodation services, the manual management system used at Pearls of Wisdom Hostel creates several challenges for both students and hostel administrators:

Lack of Real-Time Availability Information: Students often have to visit the hostel physically to confirm room availability. This can lead to wasted time and transport costs, especially if rooms are already fully booked.

Long Queues and Delays: During peak booking periods, such as the beginning of semesters, many students arrive at the hostel at the same time, causing long queues and delays in service.

Overbooking and Booking Conflict: Manual record-keeping increases the risk of errors, which may lead to double bookings or situations where more rooms are promised than are actually available due to poor Communication since students may not receive timely updates regarding room availability, maintenance schedules, hostel rules, or payment deadlines.

Inefficient Rent Collection and Record-Keeping: Heavy reliance on cash payments and handwritten records increases the likelihood of lost receipts, inaccurate records, and difficulties in tracking rent payments or identifying students who have not paid.

Administrative Burden for Hostel Management: Manual record systems make it difficult for managers to produce accurate reports on room occupancy, financial collections, maintenance needs, or booking trends, which affects planning and decision-making.

How an Online Hostel Management System Will Address These Issues

To improve efficiency and service delivery, Pearls of Wisdom Hostel can implement an Online Hostel Management System that digitizes the booking and management process. Students will be able to register or log into the system using their email address, phone number, or identification details. Through the platform, they will search for available rooms based on filters such as room type, price range, amenities, and availability dates.

The system will display real-time room availability, room descriptions, photos, and details about facilities provided by the hostel. After selecting a room, students will enter their personal information, review the booking details, and make secure payments using mobile money services such as MTN MoMo or Airtel Money.

Once payment is completed, the system will automatically generate an electronic receipt (e-receipt) with a unique booking code or QR code, which will be used for quick verification during check-in. Students will also have access to a personal dashboard where they can view booking history, payment status, and submit maintenance requests.

For hostel administrators, the system will provide a comprehensive management dashboard that allows them to add and manage rooms and hostel facilities, update room status (available, booked, occupied, or under maintenance), track student bookings and payments, send notifications and reminders to students, and generate reports on occupancy, revenue, and payment records.

By implementing this Online Hostel Management System, Pearls of Wisdom Hostel will significantly improve transparency, efficiency, and communication. The system will reduce booking conflicts, simplify payment management, minimize administrative workload, and enhance the overall accommodation experience for students living in Fort Portal.

3. Problem Statement

Pearls of Wisdom Hostel currently uses a manual booking system, which causes inaccurate room availability, long queues during peak periods, booking errors, and communication gaps. This leads to student frustration, wasted time, and administrative challenges, including poor record keeping, delayed payments, and possible revenue losses hence need for an Online Hostel Management System which will improve efficiency by providing real-time room availability, secure mobile money payments, electronic receipts, automated notifications, and

better record management, helping to reduce queues, prevent overbooking, and improve overall hostel operations.

4. Objectives

4.1. Main Objective

To develop an Online Hostel Management System.

4.2. Specific Objectives

- I. To gather user requirements
- II. To Design the New Online Hostel Management System
- III. To Develop the New Online Hostel Management System
- IV. To Test and Evaluate the New Online Hostel Management System

5. Methodology

5.1. Data collection

Data collection is the process of gathering and measuring information from relevant sources in a systematic and organized manner in order to answer research questions and make informed decisions about a study.

5.1.1. Interviews

An interview is a formal or informal conversation between the researchers and an interviewee in which the researchers asks questions to gather information from the interviewee. The researchers will physically visiting the manger and the students, asking them open-ended questions using a prepared interview guide, and recording their responses with consent. The recorded data will then be analyzed to identify user requirements that will guide the design of the hostel management system because they allow the researchers to gather detailed, in-depth information directly from key stakeholders about their real experiences and challenges with the current hostel management process.

5.1.2. Questionnaires

Questionnaire is a research instrument consisting of a set of written questions designed to collect information from a large number of respondents quickly and efficiently. The researchers will design questionnaires with a mix of closed and open-ended questions and distributed to a large sample of MMU students and hostel staff either physically or digitally via tools like Google Forms because they allow the researchers to collect data from a large number of MMU students and hostel staff quickly and efficiently without requiring physical presence at every location and its a cost-effective and time-saving way to gather measurable data on students' experiences and satisfaction levels with the current hostel booking and payment process, which can then be compared and analysed to support the development of the proposed system.

5.2. Design

The design phase is critical in establishing a user centered approach for the online hostel management system. This includes:

5.2.1. Paper prototyping

Paper prototyping is a design technique where rough, hand-drawn sketches of a system's user interface are created on paper to simulate how the actual system will look before any real coding or development begins. The researchers will create paper based sketches of the system's key screens, testing them with selected MMU students and hostel staff to gather feedback, and then refining the designs based on the feedback received before actual development begins, in a way that it provides a cheap, quick, and risk-free way to visualise and test the proposed hostel management system's design with real users before investing time and resources into actual development, allowing the researchers to identify and correct usability problems and design flaws at an early stage.

5.3. Development

In the development phase, the focus shifts to building the Online Hostel Management System using a structured approach known as Modular Development.

5.3.1 Modular Development:

This methodology involves breaking down the entire system into smaller, reusable components which are developed individually after they are integrated together to form system.

The hostel management system will be broke into distinct functional modules such as login, room booking, and payment processing, developing and testing each module independently, and then progressively integrating them into a complete and fully functional system because it simplifies the development of the complex hostel management system by breaking it into smaller, manageable, and independently developed components, making it easier to detect and fix errors, update individual features, and ensure that each part of the system functions correctly before being combined into a complete and reliable system.

5.4. Testing

The testing phase is vital for ensuring the reliability, performance, and user satisfaction of the Online Hostel Management System. Several methodologies will be employed to assess different aspects of the system; these include:

5.4.1. Unit Testing

Unit testing is a software testing method where individual components of a system such as a single function, module, or feature are tested in isolation to verify that each one works correctly. The researchers will test individual component of the hostel management system, such as the login, room booking, and payment features, in isolation to verify that each one functions correctly before the complete system is assembled and tested because it allows the researchers to detect and fix bugs and errors in each individual component of the hostel management system early during the development phase, before they spread and affect other parts of the system

5.4.2. Integration Testing

Integration testing is where individual components of a system that have already been unit tested are combined and tested together as a group to verify that they work correctly and communicate with each other as expected. The individually tested components of the hostel management system will be combined and test the combined system to verify that the components interact, communicate, and exchange data correctly as a complete system because even when individual components of the hostel management system pass unit testing, errors and defects can still arise when those components are combined and interact with each other,

so integration testing ensures that all modules such as login, room booking, and payment processing work together seamlessly and exchange data correctly to deliver a fully functional and reliable system.